

Contrast All The Time: Learning Time Series Representation from Temporal Consistency

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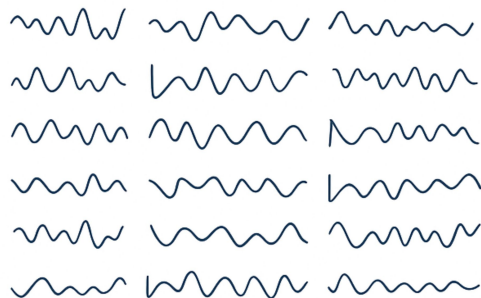
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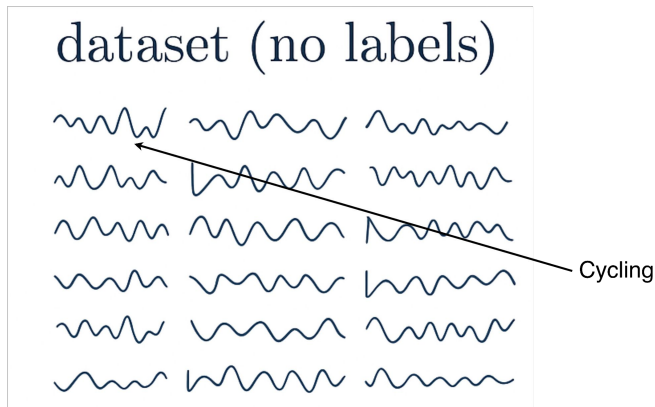
European Conference on Artificial Intelligence (ECAI 2025)

Self-supervised learning motivation

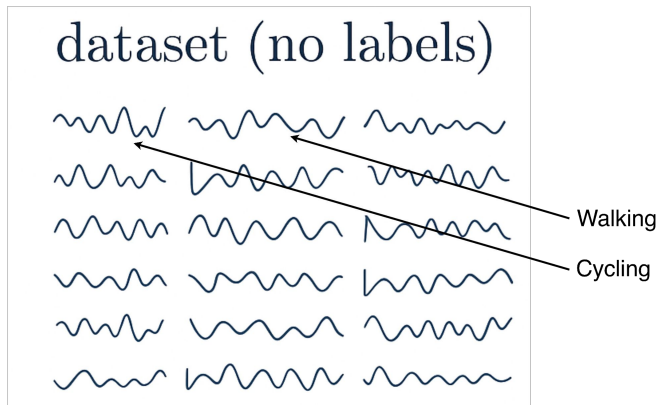
dataset (no labels)



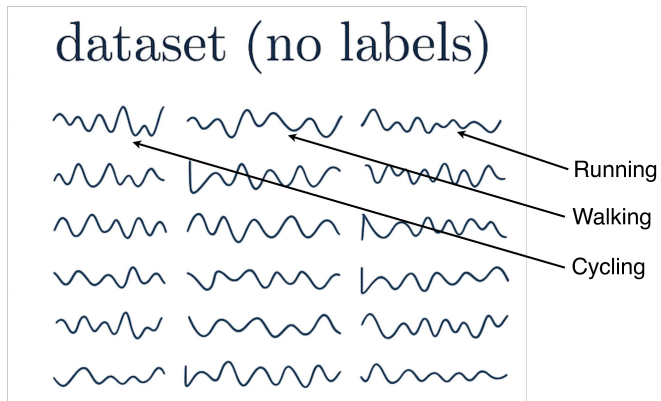
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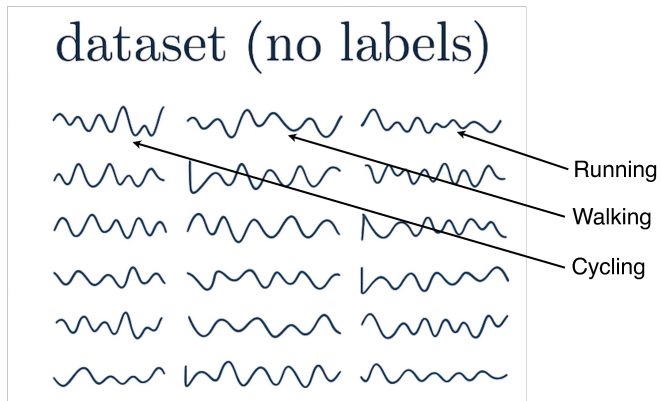
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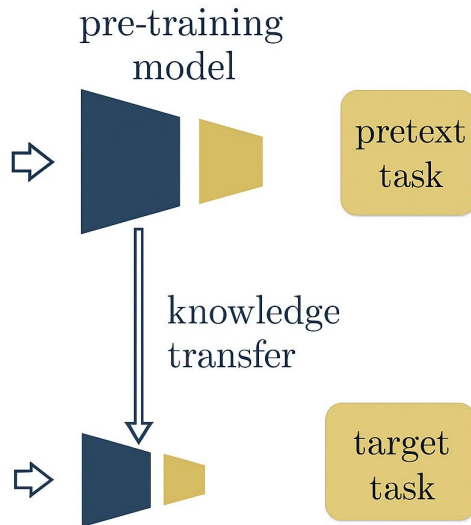
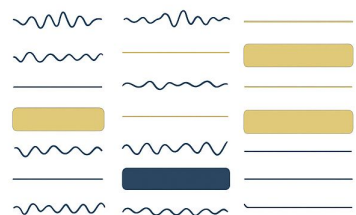
- Supervised learning
 - Relies on large labeled datasets
 - Expensive and domain-specific annotations
 - Limited generalization

Self-supervised learning motivation

dataset (no labels)

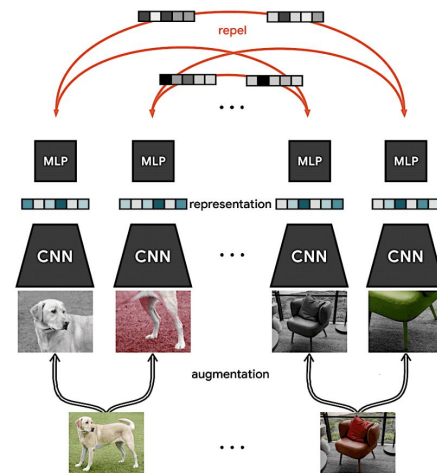
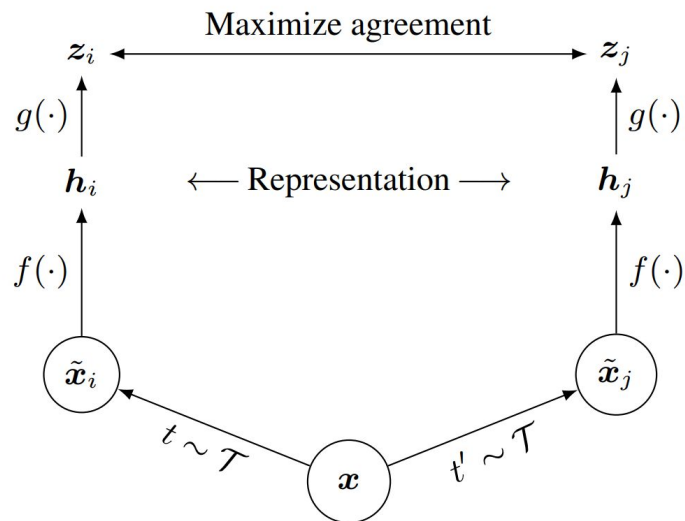


dataset (with labels)

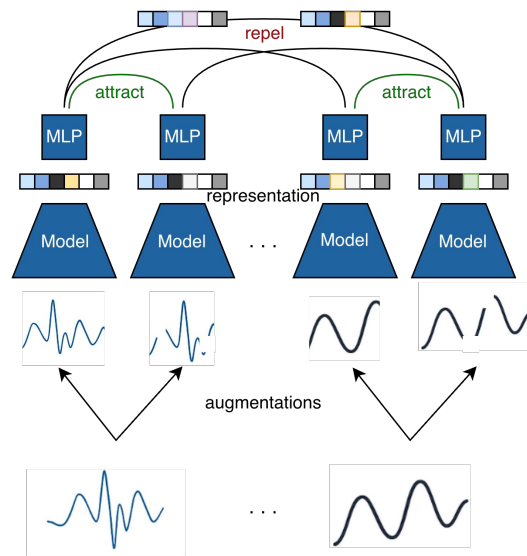
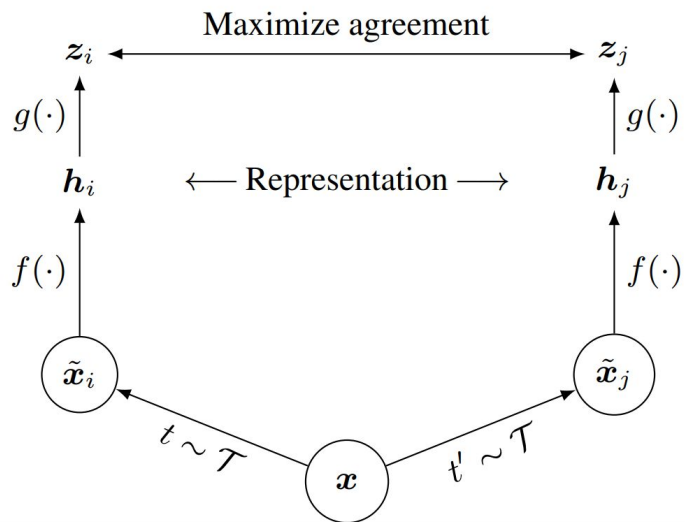


- Supervised learning
 - Relies on large labeled datasets
 - Expensive and domain-specific annotations
 - Limited generalization
- Self-supervised learning
 - Learns representations directly from data
 - Scalable (unlabeled data is abundant)
 - Improves transferability to downstream tasks

Contrastive representation learning



Contrastive representation learning



Contrastive learning in time series

TNC¹, MF-CLR², CoST³, InfoTS⁴, TS2Vec⁵

- Augmentations and masking
- Dynamic Time Warping (DTW)
- Multiple models to capture granularity

Consequence

- Data distortion
- Scalability and speed

[1] Unsupervised Representation Learning for Time Series with Temporal Neighborhood Coding, ICLR 2021

[2] MF-CLR: Multi-Frequency Contrastive Learning Representation for Time Series, ICML 2024

[3] CoST: Contrastive Learning of Disentangled Seasonal-Trend Representations for Time Series Forecasting, ICLR 2022

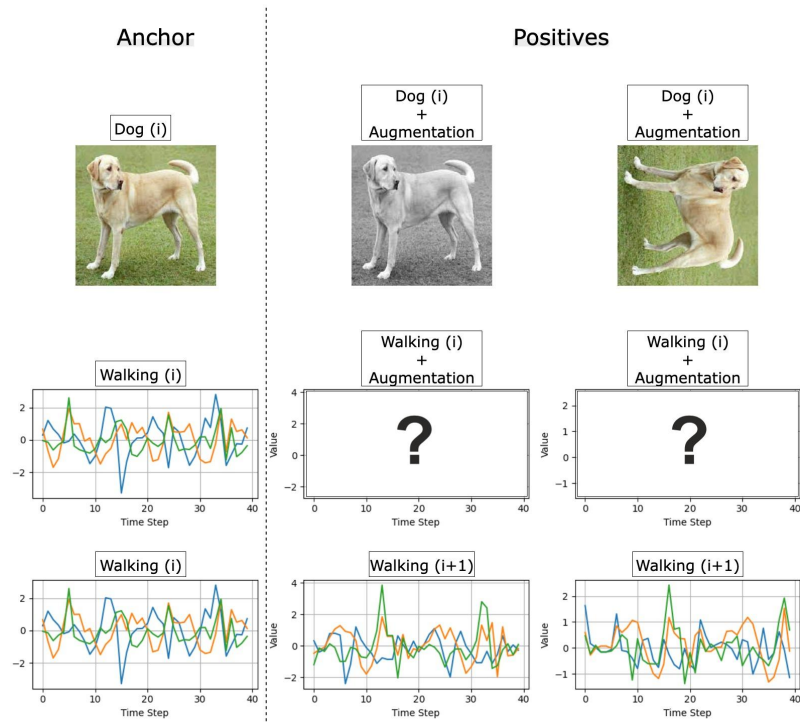
[4] Time Series Contrastive Learning with Information-Aware Augmentations, AAAI 2023

[5] TS2Vec: Towards Universal Representation of Time Series, AAAI 2022

Looking at the temporal information

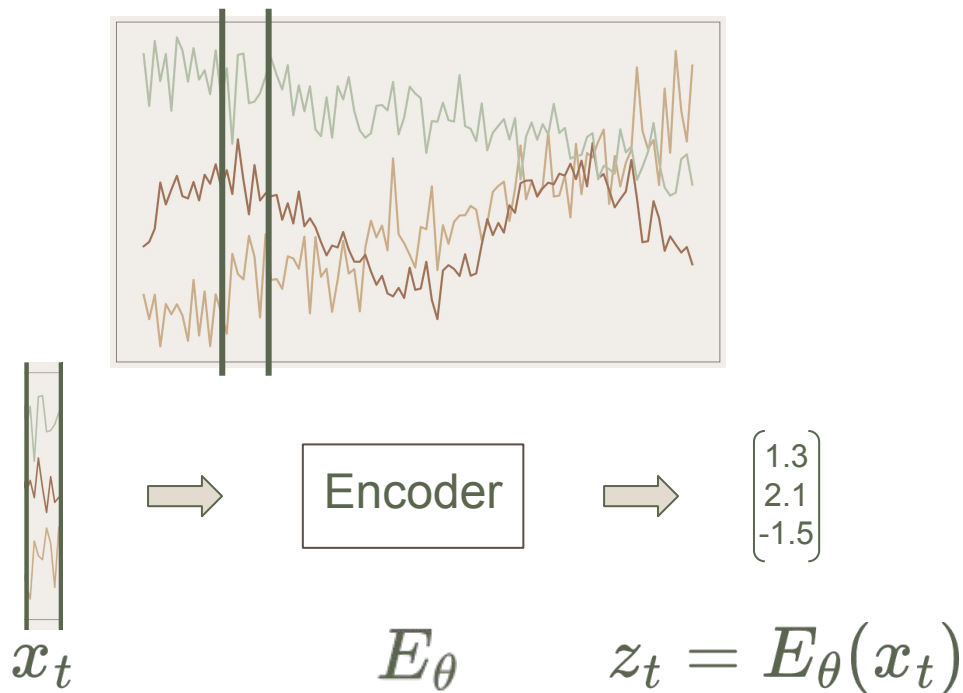
Similarity within nearby instances

- That consecutive instances in a sequence have the same class and event labels would not change too often
- Often holds for time series, which have repeated labels in the temporal dimension



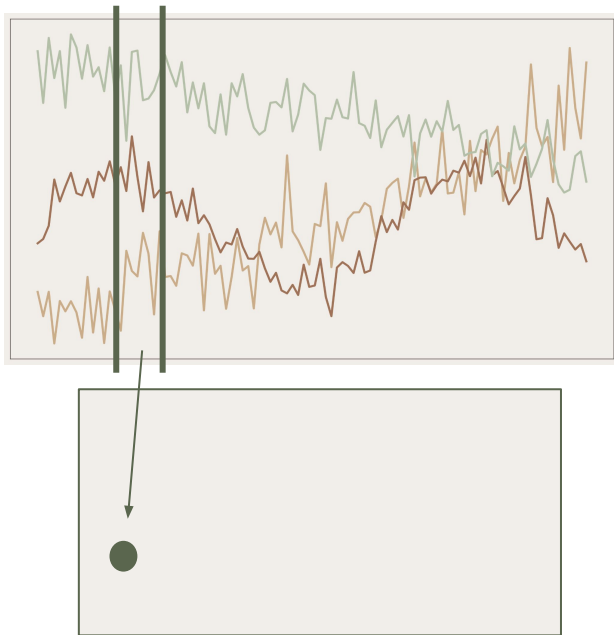
Looking at the temporal information

Learn an embedding for small segments (instances) of time series in an unsupervised way so things that appear similar are close in some space



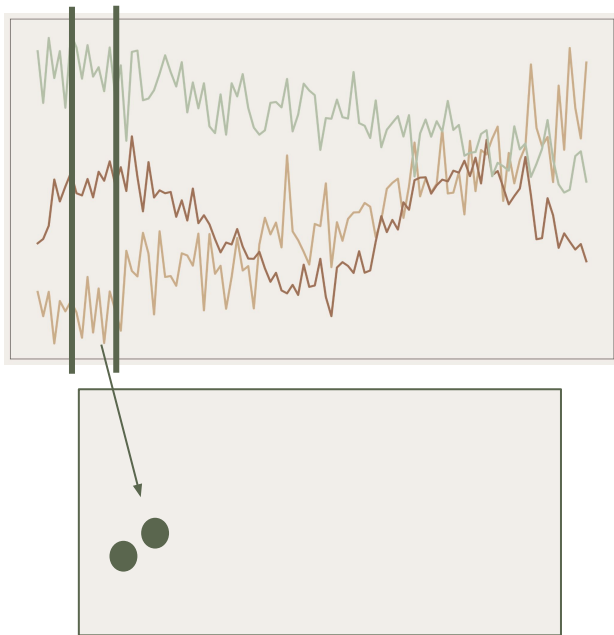
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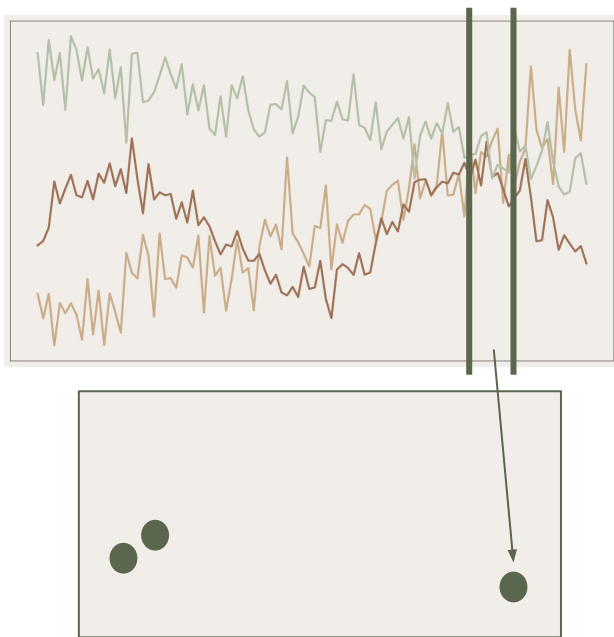
Looking at the temporal information

Learn an embedding for small segments (instances) of time series in an unsupervised way so things that appear similar are close in some space



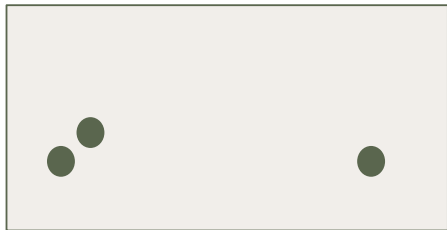
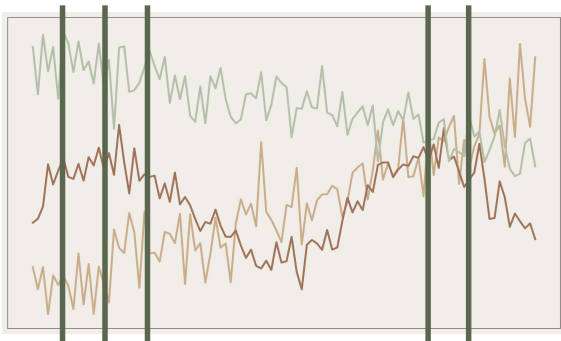
Looking at the temporal information

Learn an embedding for small segments (instances) of time series in an unsupervised way so events that appear similar are close in the embedding space



Looking at the temporal information

- Discard Augmentation
- Pattern within the data (Similarity within nearby instances)
- Scalable ?



Contrast everything!

Contrast All The Time (CaTT)!

N-pair tuple loss - Multiple positive

Positive pairs

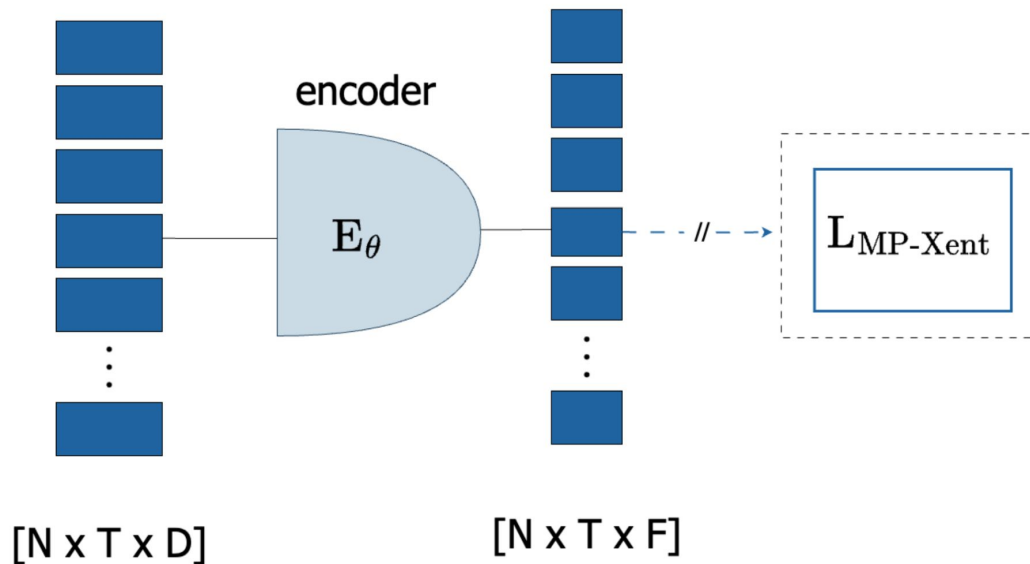
$$\ell(i) = -\log \frac{\exp(\text{sim}(z_i, z_{i-1})/\tau) + \exp(\text{sim}(z_i, z_{i+1})/\tau)}{\sum_{k=1}^N \mathbf{1}_{[k \neq i, i-2]} \exp(\text{sim}(z_i, z_{k+1})/\tau) + \sum_{l=1}^N \mathbf{1}_{[l \neq i, i-1]} \exp(\text{sim}(z_{i-1}, z_l)/\tau)}$$

Negative pairs

	Instance 1	Instance 2	Instance 3	Instance 4	Instance 5	Instance 6
Instance 1		sim(1, 2)	sim(1, 3)	sim(1, 4)	sim(1, 5)	sim(1, 6)
Instance 2	sim(2, 1)		sim(2, 3)	sim(2, 4)	sim(2, 5)	sim(2, 6)
Instance 3	sim(3, 1)	sim(3, 2)		sim(3, 4)	sim(3, 5)	sim(3, 6)
Instance 4	sim(4, 1)	sim(4, 2)	sim(4, 3)		sim(4, 5)	sim(4, 6)
Instance 5	sim(5, 1)	sim(5, 2)	sim(5, 3)	sim(5, 4)		sim(5, 6)
Instance 6	sim(6, 1)	sim(6, 2)	sim(6, 3)	sim(6, 4)	sim(6, 5)	

Contrast All The Time (CaTT)

MP-Xent NT tuple loss per update (efficient training)



Downstream tasks - Classification

ECG

Normal heartbeat



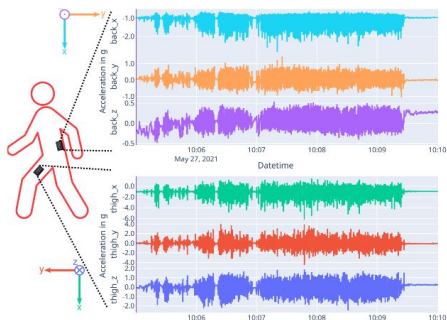
Atrial flutter



Atrial fibrillation

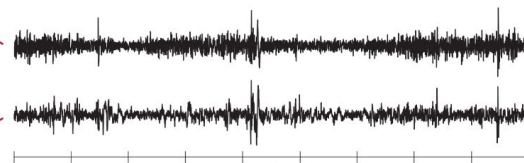
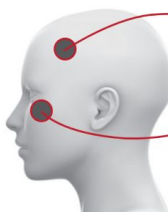


HARTH

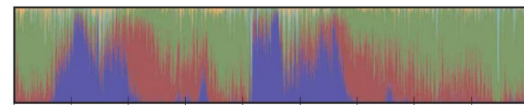


	# Instance	Sequence length	Dimension	Classes
HARTH	1,270,087	119	156	12
SLEEP	478,600	200	90	5
EEG	255,200	200	1502	4

SleepEEG



High-Frequency Sleep Stage Representation
128 Hz



Segment Classification
E.g. 1/30 Hz



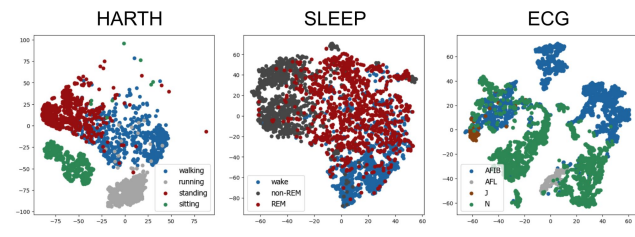
Linear evaluation with frozen backbone

Datasets	Models	Accuracy	F1 score	Precision	Recall	Time (s) ↓
HARTH	Random Init.	86.69±0.50	0.84±0.01	0.55±0.01	0.45±0.02	-
	TNC	89.03±0.03	0.88±0.00	0.70±0.02	0.55±0.01	13.86
	InfoTS	90.12±0.13	0.89±0.00	0.71±0.02	0.56±0.01	186.50
	CoST	88.68±0.41	0.88±0.00	0.64±0.01	0.56±0.00	12.61
	SimMTM	68.14±4.83	0.60±0.08	0.34±0.04	0.24±0.05	245.67
	TimeDRL	71.92±2.37	0.63±0.04	0.37±0.04	0.28±0.01	20.60
	TS2Vec	87.59±0.23	0.85±0.00	0.58±0.01	0.49±0.00	34.96
	TS2Vec + <i>SoftCLT</i>	88.07±0.13	0.86±0.00	0.62±0.03	0.50±0.01	37.65
	MF-CLR	77.07±5.38	0.73±0.07	0.52±0.04	0.37±0.05	2322.75
	CaTT	90.35±0.05	0.89±0.00	0.71±0.02	0.59±0.00	6.12
SLEEP EEG	Random Init.	46.89±11.34	0.44±0.08	0.36±0.03	0.32±0.04	-
	TNC	55.13±0.50	0.54±0.01	0.33±0.01	0.36±0.02	10.14
	InfoTS	43.32±9.66	0.42±0.07	0.41±0.05	0.35±0.03	260.35
	CoST	56.63±2.56	0.55±0.00	0.38±0.02	0.35±0.04	11.89
	SimMTM	57.08±0.68	0.50±0.01	0.31±0.03	0.25±0.01	1739.66
	TimeDRL	57.30±2.22	0.48±0.02	0.36±0.09	0.24±0.02	13.20
	TS2Vec	51.18±1.12	0.50±0.00	0.29±0.01	0.35±0.02	34.42
	TS2Vec + <i>SoftCLT</i>	53.60±2.44	0.52±0.01	0.34±0.03	0.37±0.02	36.48
	MF-CLR	57.07±1.89	0.48±0.02	0.27±0.06	0.27±0.01	2318.26
	CaTT	62.39±1.21	0.60±0.01	0.47±0.00	0.41±0.01	6.36
ECG	Random Init.	65.48±1.44	0.64±0.01	0.35±0.01	0.34±0.01	-
	TNC	75.86±2.00	0.75±0.02	0.39±0.00	0.38±0.01	4.86
	InfoTS	61.94±3.19	0.59±0.05	0.34±0.01	0.32±0.01	76.42
	CoST	69.37±9.55	0.67±0.13	0.38±0.02	0.36±0.04	11.27
	SimMTM	71.89±2.27	0.71±0.03	0.37±0.00	0.36±0.01	594.86
	TimeDRL	59.51±6.93	0.55±0.10	0.33±0.02	0.31±0.03	13.21
	TS2Vec	77.72±1.27	0.77±0.01	0.40±0.00	0.39±0.01	11.42
	TS2Vec + <i>SoftCLT</i>	76.37±3.88	0.76±0.04	0.39±0.02	0.39±0.02	16.70
	MF-CLR	52.69±2.29	0.45±0.06	0.31±0.01	0.28±0.01	3056.11
	CaTT	80.18±3.80	0.81±0.03	0.41±0.01	0.41±0.02	4.47

CaTT delivers state-of-the-art performance

Fastest computational time

t-SNE plots of the learned embeddings



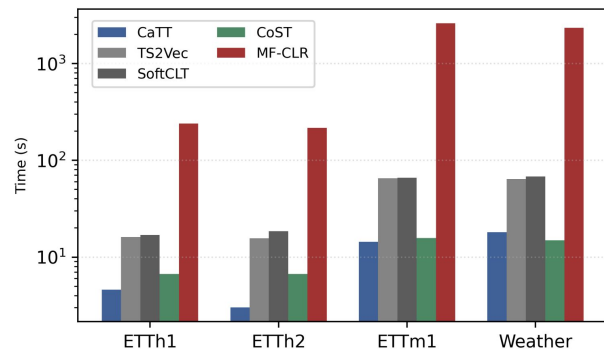
Short and long horizon forecasting

Dataset	H	CaTT		TS2Vec		TS2Vec + <i>SoftCLT</i>		CoST		MF-CLR	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh ₁	24	0.040	0.152	0.040	0.153	0.040	0.153	0.041	0.152	0.039	0.149
	48	0.063	0.193	0.063	0.193	0.063	0.192	0.066	0.196	0.062	0.189
	168	0.122	0.269	0.145	0.296	0.144	0.296	0.126	0.273	0.142	0.297
	336	0.143	0.296	0.164	0.321	0.164	0.321	0.150	0.305	0.165	0.323
	720	0.189	0.354	0.174	0.340	0.174	0.342	0.188	0.354	0.201	0.371
ETTh ₂	24	0.095	0.235	0.090	0.227	0.089	0.226	0.091	0.231	0.093	0.230
	48	0.130	0.279	0.126	0.275	0.125	0.274	0.127	0.277	0.133	0.279
	168	0.190	0.348	0.200	0.357	0.201	0.357	0.206	0.359	0.200	0.352
	336	0.196	0.357	0.205	0.365	0.206	0.366	0.206	0.363	0.206	0.363
	720	0.201	0.365	0.205	0.370	0.205	0.370	0.205	0.368	0.202	0.364
ETTM ₁	24	0.015	0.092	0.016	0.093	0.014	0.089	0.014	0.087	0.015	0.090
	48	0.028	0.124	0.034	0.137	0.027	0.123	0.027	0.121	0.028	0.125
	96	0.044	0.159	0.049	0.169	0.045	0.162	0.044	0.158	0.043	0.156
	288	0.086	0.225	0.099	0.252	0.097	0.238	0.092	0.233	0.094	0.234
	672	0.124	0.271	0.149	0.296	0.144	0.289	0.137	0.289	0.134	0.280
Weather	24	0.037	0.152	0.131	0.317	0.105	0.276	0.049	0.198	0.080	0.257
	48	0.024	0.114	0.094	0.226	0.077	0.202	0.034	0.153	0.049	0.181
	168	0.012	0.082	0.034	0.120	0.030	0.118	0.013	0.080	0.017	0.086
	336	0.009	0.069	0.021	0.097	0.019	0.094	0.011	0.074	0.012	0.074
	720	0.011	0.083	0.015	0.086	0.014	0.084	0.009	0.070	0.011	0.075
Avg.		0.088	0.211	0.103	0.235	0.099	0.229	0.092	0.217	0.096	0.224

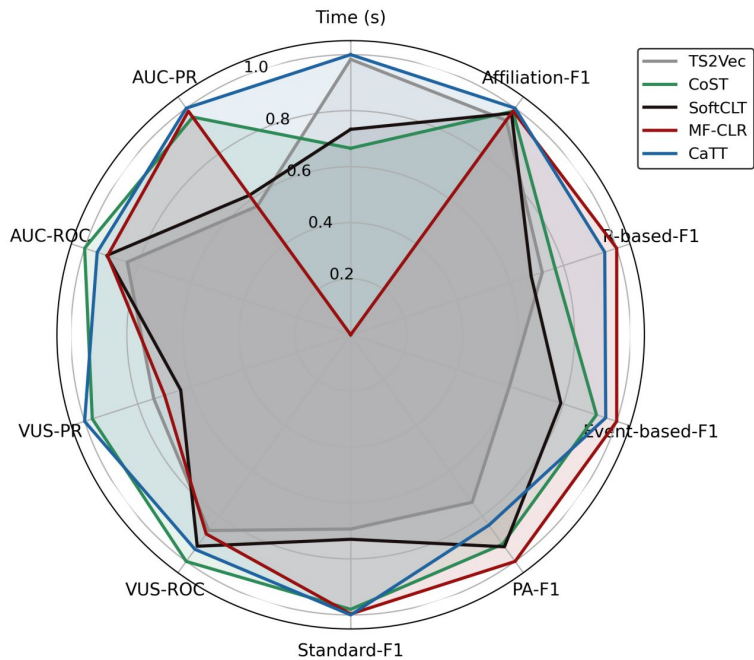
Timestamp level resolution

Overall CaTT achieves the lowest prediction MSE

Lowest computational time



Additional experiment



TS representation learning methods

Nine multivariate anomaly detection datasets - TSB-AD

Lowest computational time

What did we learn?

- Focusing on temporal information rather than augmentation might be the way forward
 - Demonstrates the ability to learn semantically meaningful representations
 - Learns timestamp-level representation, as underscored by its performance in time series forecasting

What did we learn?

- Focusing on temporal information rather than augmentation might be the way forward
 - Demonstrates the ability to learn semantically meaningful representations
 - Learns timestamp-level representation, as underscored by its performance in time series forecasting
- Avoiding augmentations, DTW and multiple models to learn representation, speeds up training
 - Delivers top performance in downstream classification and forecasting tasks with the shortest training time

Paper:

<https://arxiv.org/pdf/2410.15416>



Code:

<https://github.com/sfi-norwai/CaTT>

